3. Project Planning Phase

3.1 Introduction

The project planning phase establishes the objectives, scope, resources, activities, schedule, and risks of EduGenie. Its purpose is to create a clear roadmap so that development activities are organized, resources are used effectively, and the project is completed within the specified time.

3.2 Project Objectives

- To develop an easy-to-use AI-powered learning assistant.
- To provide secure user registration and authentication.
- To integrate Google Gemini for explanations, answers, and content generation.
- To enable notes summarization, quiz generation, and flashcards.
- To generate personalized study plans.
- To maintain learning history and display progress through a dashboard.
- To help learners understand topics faster and revise more effectively.

3.3 Project Scope

The scope includes the design and development of a web-based learning assistant. Users can create an account, log in, chat with the AI tutor, request explanations, summarize notes, take generated quizzes, review flashcards, create study plans, and view their learning history and progress. Mobile apps, voice input, and offline mode are outside the current scope and are listed as future enhancements.

3.4 Project Modules

- Authentication Module
- AI Tutor Chat Module
- Concept Explanation Module
- Notes Summarization Module
- Quiz Module
- Flashcard Module
- Study Plan Module
- History and Dashboard Module

3.5 Resource Planning

Resources include the development computer, internet connection, Python and FastAPI, a web frontend, a relational database, a Google Gemini API key, Visual Studio Code, and Git/GitHub for version control.

3.6 Project Schedule

Phase	Activities	Duration
1	Brainstorming, problem identification, idea finalization	Week 1
2	Requirement analysis and documentation	Week 1–2
3	System design, architecture, database design	Week 2–3
4	Backend and frontend development, Gemini integration	Week 3–6
5	Testing and bug fixing	Week 6–7

Phase	Activities	Duration
6	Demonstration preparation and final documentation	Week 7–8

3.7 Risk Analysis

Risk	Impact	Mitigation
Gemini API downtime or rate limits	AI features temporarily unavailable	Error handling, retry logic, user-friendly messages
Inaccurate AI responses	Learner may receive incorrect information	Clear prompts, disclaimer, encourage verification
API key exposure	Security and misuse risk	Store key in server environment variables
Schedule delays	Incomplete features	Prioritize core modules first
Slow responses	Poor user experience	Loading indicators, concise prompts